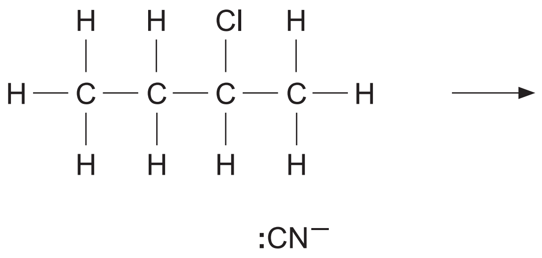
Examiner only

10. *(a)* Halogenoalkanes react with potassium cyanide in a similar way to their reaction with $OH^-(aq)$.

(i) Complete the mechanism of the reaction between 2-chlorobutane and potassium cyanide. You should include relevant dipoles and curly arrows to show the movement of electrons. [4]



(ii) Name the type of reaction mechanism shown in part (i). [1]

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(b) When 2-chlorobutane is heated with sodium hydroxide dissolved in ethanol three different organic products are formed.

(i) Draw the displayed formulae of the **three** products. Name each product. [3]

(ii) Name the type of reaction taking place. [1]

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(c) Until recently chlorofluorocarbons, CFCs, were used in many commercial products but nowadays hydrofluorocarbons, HFCs, are preferred.

Explain why HFCs are preferred to CFCs. You do **not** need to include the mechanism or equation for any reaction which you describe. [5]

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